Course: BM20A4703 Partial Differential Equations with Applications.
Lecturer: Duc-Lam Duong (duc-lam.duong@lut.fi). TA: Fabian Schneider (fabian.schneider@lut.fi)
Due on: **17/02/2025 23:59**

Worksheet 5

Exercise 5.1

[20] Solve the following initial-boundary value problems using the method of separation of variables (write down the solution explicitly in the sine series)

$$u_{tt} - u_{xx} = 0, \quad x \in [0,1],\ t > 0,$$

with $u(x,0) = f(x),\ u_t(x,0) = 0,\ u(0,t) = u(1,t) = 0$, where

$$f(x) = \begin{cases} x, & \text{if } 0 \le x \le \frac{1}{2}, \\ 1-x, & \text{if } \frac{1}{2} \le x \le 1. \end{cases}$$

Exercise 5.2

[20] Using the formula given by the method of separation of variables (Theorem 4.2), prove that the solution of the wave equation $u_{tt} - u_{xx} = 0,\ x \in (0, 2\pi)$, is stable with respect to changes in initial data. That is, there exists a constant $C > 0$ such that

$$\sup_{t>0} \|u_1 - u_2\|_{L^2}^2 \le C \left(\|f_1 - f_2\|_{L^2}^2 + \|g_1 - g_2\|_{L^2}^2 \right),$$

where u_1, u_2 are solutions of the wave equation w.r.t Cauchy initial conditions $f_1, g_1 \in L^2(0, 2\pi)$ and $f_2, g_2 \in L^2(0, 2\pi)$, respectively.
Hint: Use Theorem 4.2 and the fact that $(w_k(x))_{k\in\mathbb{N}}$ form a complete orthonormal basis of $L^2(0, 2\pi)$, you can write

$$u = \sum_{k=1}^{\infty} \langle u, w_k \rangle w_k,$$

for any $u \in L^2(0, 2\pi)$.

Exercise 5.3

[20] Let u solve the wave equation $u_{tt} - \Delta u = 0$ in $\mathbb{R}^n \times (0, \infty)$ with initial conditions $u(x,0) = f(x), u_t(x,0) = g(x)$. Assume that $f, g, \nabla f \in L^2(\mathbb{R}^n)$. Denote $E_K(t), E_P(t)$ to be the kinetic and potential energies of u, that is,

$$E_K(t) = \frac{1}{2} \int_{\mathbb{R}^n} u_t^2 \mathrm{d}x, \quad E_P(t) = \frac{1}{2} \int_{\mathbb{R}^n} |\nabla u|^2 \mathrm{d}x.$$

Prove that

$$\lim_{t\to\infty} E_P(t) = \lim_{t\to\infty} E_K(t).$$

Hint: First notice that the total energy $E(t) = E_K(t) + E_P(t)$ is conserved. Use the Fourier transform (in particular, equation (4.24) in Section 4.4 and Plancherel's identity (Theorem B.6 in Appendix B) to prove that

$$\lim_{t\to\infty} E_P(t) = \frac{1}{4} \int_{\mathbb{R}^n} |\nabla f|^2 + |g|^2 \mathrm{d}x = \frac{1}{2} E(0),$$

which is one-half of the total energy.

©Duc-Lam Duong